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CZSRSSC: CONTINUAL ZERO-SHOT REMOTE SENSING SCENE CLASSIFICATION

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ABSTRACT

Remote sensing scene classification, a core technology in fields such as disaster response, resource management, and urban planning, often faces challenges in real-world applications, such as identifying unlabeled new classes (zero-shot learning) and dynamically expanding classes (continuous learning). While continual zero-shot scene classification is actively researched in common scenarios, it remains largely unresolved in remote sensing scenarios. Furthermore, a major issue currently hindering the continued development of zero-shot scene classification in remote sensing is the lack of specialized datasets. To address these issues, we conducted the following research: First, we introduced continual zero-shot scene classification into the remote sensing field for the first time, breaking the traditional remote sensing scene classification's reliance on predefined classes of all classes and achieving effective classification of dynamically increasing remote sensing scene classes. Secondly, we established the first dedicated remote sensing dataset for continual zero-shot scene classification tasks by integrating multiple public remote sensing datasets, constructing unified semantic information, selecting high-quality samples, and extracting image and attribute features. Finally, we verified the effectiveness of the existing continual zero-shot scene classification method in common scenes in remote sensing scenarios based on the RSCZSL dataset, and compared it with the common zero-shot scene classification method.

***Index Terms*—** Zero-shot learning, Continual learning, Remote sensing, Dataset, Continual zero-shot remote sensing scene classification.

1. INTRODUCTION

Zero-Shot Remote Sensing Scene Classification (ZSRSSC) [1] learns the relationship between image features and attribute features of seen classes, enabling it to accurately classify unseen class images by virtue of the previously provided attribute features when such images appear. With the rapid development of earth observation and deep learning technologies, ZSRSSC has made significant contributions to fields such as dynamic environmental monitoring, disaster emergency response, urban planning, and smart town management [2, 3, 4, 5]. Meanwhile, the continuous emergence of new technologies and new data has also promoted the rapid development of ZSRSSC towards high precision.

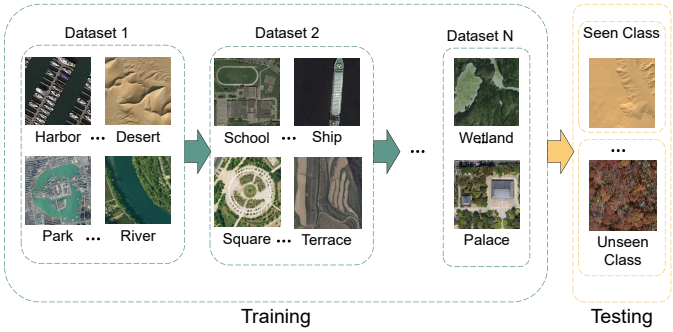


Fig. 1. Continual zero-shot scene classification for remote sensing: Whenever a new remote sensing dataset arrives, we continue to learn the features of the new dataset without forgetting the knowledge learned from the previous dataset, and eventually we are able to classify all classes of all datasets (including seen and unseen classes).

Zero-Shot Scene Classification has undergone an evolutionary process, progressing from Embedding-based Zero-Shot Scene Classification [6, 7, 8] to Generative-based Zero-Shot Scene Classification [9, 10, 11], and then to Large Model-based Zero-Shot Scene Classification [12, 13, 14]. Different from traditional zero-shot learning, which only acquires image features and attribute information from a single dataset, continual zero-shot learning can learn knowledge sequentially from multiple datasets with significant differences [15]. Meanwhile, based on the attribute information of unseen classes provided by these datasets, it can accurately identify and classify all images belonging to unseen classes—this capability breaks the dependence of traditional zero-shot scene classification on "full-category predefinition".

At the same time, with the rapid development of carrier rockets and earth observation technologies, a large amount of unannotated remote sensing image data has emerged [16,

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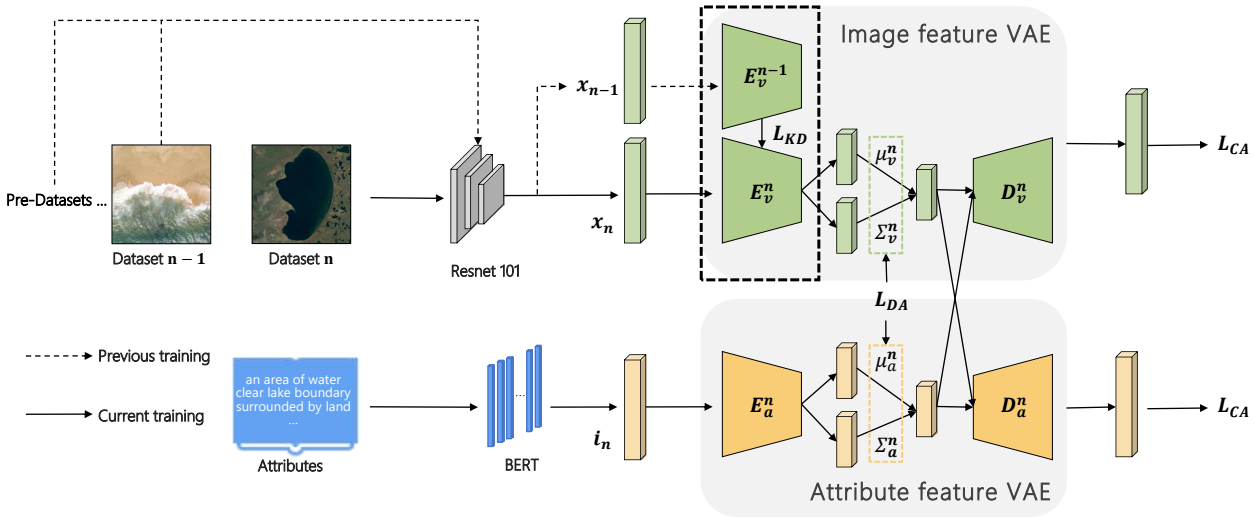


Fig. 2. The architecture of the CADA-VAE continuous zero-shot scene classification method.

17]. Compared with ordinary images, manual annotation of remote sensing images is more time-consuming and labor-intensive due to factors such as inter-class similarity and intra-class diversity. Meanwhile, as the precision of earth observation continues to improve, more detailed class division leads to the constant emergence of new classes—this inherently necessitates the integration of zero-shot scene classification and continual learning techniques, as shown in Fig.1. However, despite the enormous volume of remote sensing images, the number of remote sensing datasets available for zero-shot scene classification is extremely scarce; currently, only RSSDIVCS [18] exists. Moreover, in the field of continual zero-shot scene classification, remote sensing datasets remain a complete blank. To address this gap, we have created the first remote sensing continual zero-shot scene classification dataset, RS-CZSL. This dataset integrates content from four other public remote sensing datasets (including RSSDIVCS), and we have conducted unified construction of category attribute information and feature extraction for it.

For continual zero-shot scene classification, generative methods have an inherent advantage, as already trained generative models can generate training set content from previous datasets when new datasets arrive. After mixing in the new datasets, they train together, and finally, the generative model trains the classifier, resulting in a new model that can accommodate all features from both new and old datasets. Based on this, [10] proposed a generative method that combines selective retraining and knowledge distillation. This method uses cross-distribution aligned VAE to encode and decode image features and semantic information, and trains the classifier based on the trained VAE model. To prevent overfitting, certain parameters are selectively frozen during each training session to prevent the model from completely forgetting the knowledge learned from previous datasets. Additionally, various loss functions are employed to constrain the model's training. Extensive experiments show that this method for continual zero-shot scene classification outperform standard ZSL methods in the remote sensing field, but accuracy still sees a significant drop compared to common scene datasets. The innovations of this paper are as follows:

- Within the scope of our research, we are the first to introduce continual zero-shot scene classification into the field of remote sensing images.
- We have released a large-scale dataset(RS-CZSL) for continual zero-shot scene classification of remote sensing images, which can serve as a benchmark for continual zero-shot remote sensing scene classification and facilitate subsequent research on this field.
- A large number of experiments were conducted using the RS-CZSL on existing zero-shot and continual zero-shot scene classification methods, verifying the effectiveness of the dataset and the current continual zero-shot scene classification methods.

2. METHODOLOGY

In order to introduce continual zero-shot scene classification into the field of remote sensing, we adopted a state-of-the-art continual zero-shot scene classification method. This method utilizes the VAE model to map the features of different datasets to a shared latent space, then employs a selective retraining strategy to prevent overfitting caused by excessive training and negative transfer, and finally leverages knowledge distillation techniques to help the model recall knowledge learned from previous tasks.

2.1. Problem Formulation

There are n datasets that arrive one after another $D_n = S_n \cup U_n$,

$$S_n \{(x_n, y_n^s, i_n) | x_n \in X_n, y_n^s \in Y_n, i_n \in I_n\},$$
$$U_n \{(u_n, y_n^u, i_n) | u_n \in X_n, y_n^u \in Y_n, i_n \in I_n\},$$

x_n represents the features of the seen class image, u_n represents the features of the unseen class image features, y_n represents the class labels, and z_n represents the attribute features. The goal of generalized continual zero-shot scene classification is to continue learning knowledge each time a new S_n arrives, while ensuring that S_{n-1} is not forgotten, ultimately being able to accurately classify all D_n.

2.2. Cross and Distribution Aligned VAE

Cross and Distribution Aligned VAE(CADA-VAE) [10] is a more advanced zero-shot scene classification method. This method includes two sets of encoders and decoders, which encode and decode the input image features and attribute features, respectively. These two types of features are mapped to the same latent space and aligned through this method, as shown in Fig.2. Within the framework of a VAE, its objective function essentially corresponds to the variational lower bound of the marginal likelihood of a given sample. The specific mathematical expression of this objective function is as follows:

$$\mathscr{L} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \lambda D_{KL}(q_\phi(z|x)||p_\theta(z)),$$

where the first term is the reconstruction loss, and the second term is the Kullback-Leibler (KL) divergence—used to regularize the inference model $q(z|x)$ and the prior distribution $p(z)$. The encoder predicts μ and Σ such that $q_\phi(z|x) = \mathscr{N}(\mu, \Sigma)$, and the latent vector is obtained via the reparameterization trick. Notably, the loss function of this model consists of three parts:

$$\mathscr{L}_{CA} = |i - D_a(E_v(x))| + |x - D_v(E_a(i))| \quad (1)$$

Cross alignment loss $\mathscr{L}_{CA}$, where x and i represent image features and attribute features respectively, E_v and D_v represent the image feature encoder and decoder, while E_a and D_a represent the attribute feature encoder and decoder. The loss of cross-alignment enhances the capability of encoders and decoders by cross-reconstructing the two types of features.

$$\mathscr{L}_{DA} = \left(\|\mu_a - \mu_v\|_2^2 + \left\| \Sigma_a^{\frac{1}{2}} - \Sigma_v^{\frac{1}{2}} \right\|_{\text{Frobenius}}^2 \right)^{\frac{1}{2}} \quad (2)$$

The loss of distribution alignment $\mathscr{L}_{DA}$, where μ_a and Σ_a are predicted by E_a, and μ_v and Σ_v are predicted by E_v. The loss of distribution alignment directly reduces the distance between the two features in the latent space through the $L2$ norm and the Frobenius norm.

$$\mathscr{L}_{VAE} = \mathscr{L}_{VAE}^a + \mathscr{L}_{VAE}^v \quad (3)$$

The VAE loss consists of two parts: the image feature VAE loss $\mathscr{L}_{VAE}^v$ and the attribute feature VAE loss $\mathscr{L}_{VAE}^a$.Finally, the total loss of the model is:

$$\mathscr{L}_{CACD-VAE} = \delta \mathscr{L}_{DA} + \gamma \mathscr{L}_{CA} + \mathscr{L}_{VAE}$$

Here, δ and γ are adjustable custom parameters.

2.3. Selective Retraining and Knowledge Distillation

When retraining a typical zero-shot model, since all parameters can be updated, after training on the new dataset, most of the parameter changes cause the model to forget the content learned from the previous dataset. However, the selective retraining strategy that we adopted effectively improved this problem. By analyzing the model weights, we select 'active neurons', mask redundant neurons, and use hook functions to freeze the gradient updates of redundant neurons, keeping only the active neurons involved in subsequent training. This method can minimize redundant parameters as much as possible, retain critical knowledge, and thus reduce the model's 'catastrophic forgetting'.

Humankind always stands on the shoulders of giants when learning knowledge, which saves a lot of time and energy. Similarly, we hope that models can do the same when learning knowledge, so we introduce the method of 'knowledge distillation' here. We use the VAE encoder trained on the previous dataset as the teacher model, which has learned the distribution rules of image features in that dataset. The student model is initialized by loading the encoder weights from the teacher model, with the goal of learning the features of the new dataset while retaining the prior knowledge from the teacher model.

3. EXPERIMENT

In this section, we will provide a detailed introduction to the RS-CZSL dataset and the experiments conducted using this dataset, in order to validate the feasibility of introducing continuous zero-shot scene classification into the field of remote sensing, as well as the effectiveness and originality of RS-CZSL.

3.1. RS-CZSL

In recent years, the development of launch vehicles and remote sensing technologies has greatly reduced the costs of ac-

Table 1. RS-CZSL Basic Information

Dataset	Images	All Classes	Seen	Unseen
RSS	56000	70	60	10
CLRS	15000	25	18	7
SIRI	2400	12	10	2
sky	8000	10	7	3

Table 2. Comparison between CADA-VAE continual zero-shot method and ordinary ZSL methods.

		sky			SIRI			CLRS			RSS		
		u	s	H	u	s	H	u	s	H	u	s	H
sky	1	0.40	0.66	0.49	-	-	-	-	-	-	-	-	-
sky+SIRI	2	0.43	0.72	0.53	0.48	0.56	0.49	-	-	-	-	-	-
sky+SIRI+CLRS	3	0.45	0.71	0.54	0.39	0.53	0.44	0.10	0.39	0.15	-	-	-
Baseline	4	**0.41**	**0.68**	**0.51**	**0.46**	**0.51**	**0.47**	**0.11**	**0.42**	**0.17**	**0.20**	**0.60**	**0.30**
Ordinary ZSL method	4	**0.33**	**0.14**	**0.2**	**0.49**	**0.1**	**0.16**	**0.14**	**0.06**	**0.08**	**0.1**	**0.75**	**0.17**
	3	0.34	0.14	0.21	0.5	0.1	0.16	0.14	0.52	0.22	-	-	-
	2	0.33	0.12	0.19	0.5	0.75	0.6	-	-	-	-	-	-
	1	0.33	0.85	0.48	-	-	-	-	-	-	-	-	-

quiring remote sensing images, leading to an increasing number of basic remote sensing datasets, such as classification, description, and segmentation. However, zero-shot datasets have more requirements compared to regular datasets: they require descriptions or attribute features for each category, as opposed to classification datasets and compared to describing datasets, it requires detailed classification. These two limitations have resulted in only one available zero-shot dataset in the remote sensing field, which is RSSDIVCS. However, continual zero-shot scene classification requires multiple zero-shot datasets to validate the cross-domain capabilities of the methods. Therefore, we constructed RS-CZSL as a benchmark dataset for continual zero-shot scene classification in the field of remote sensing. RS-CZSL integrates the content of four datasets (the abbreviation in parentheses will be used later.): RSSDIVCS (RSS) [18], CLRS [19], SIRI-WHU (SIRI) [20], and skycity-the-city-landscape (sky) [21]. We carefully selected and categorized the classes from these four datasets, with the final number of classes and images shown in Table 1. We also used Qwen2.5 to describe each class, and then employed a Resnet101 model fine-tuned on remote sensing datasets and a BERT model to extract unified image and attribute features. The RS-CZSL dataset will be publicly released soon.

3.2. Experiment Settings

Due to the lack of continual zero-shot scene classification methods in the field of remote sensing, we applied RS-CZSL to both traditional zero-shot methods and the CADA-VAE continual zero-shot scene classification method. We used the common method and the CADA-VAE continual zero-shot scene classification method to train the four datasets RSS, CLRS, SIRI, and sky simultaneously and sequentially. After completion, we performed classification tests on the test sets of the four datasets. The results are shown in bold in Table 2.

3.3. Experiment Results Analysis

In the field of remote sensing, the continual zero-shot scene classification method comprehensively surpassed the common ZSL method. On the smaller sky and SIRI datasets, the CADA-VAE continual zero-shot scene classification method improves s by nearly 50% and H by nearly 30%. On the larger CLRS and RSS datasets, the CADA-VAE continual zero-shot scene classification method also improves H by 10%. This shows that compared with large datasets, small datasets are more sensitive to changes in model parameters. This also proves the importance of introducing continuous learning into zero-shot. In reality, we usually provide the model with the largest dataset at the beginning and then provide some smaller supplementary datasets.

In addition, we save the weights of the model after each training of a data set and use these weights to test the classification of the previous dataset. It can be seen that because the model itself has a certain zero-sample classification ability, the classification accuracy of the model for unseen categories does not change much. While ordinary methods will produce "catastrophic forgetting" of the learning of categories seen in the previous dataset after each training, the CADA-VAE continual zero-shot scene classification method greatly improves the model's memory of categories seen in the previous dataset, thereby significantly improving the model's continual zero-shot learning ability.

4. CONCLUSION

Within the scope of our research, this paper is the first to apply continual zero-shot scene classification to remote sensing. First, we construct a benchmark dataset RS-CZSL for remote sensing continual zero-shot scene classification to serve as a benchmark dataset for subsequent research. Secondly, we use RS-CZSL to verify the effectiveness of existing continual zero-shot scene classification methods in common scenes in remote sensing scenarios and compare them with common zero-shot scene classification methods. Experiments have shown that the existing continual zero-shot scene classification method under common scenarios effectively alleviates the "catastrophic forgetting" of the model compared to the ordinary zero-shot method, and significantly improves the robustness of the model in identifying incremental classes.

5. ACKNOWLEDGEMENTS

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